Arbaz Siddiqui

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SUMMARY

Backend engineer with ten years of experience building consumer products at scale. At Turnip I built and ran the live-streaming and social platform serving **25M users** through four pivots (game streaming, live audio, a creator social network, and now an AI design suite), and spent the last two years on the AI infrastructure around it. Outside work I release open-weights models, build mobile apps and run AI pipelines, including a 12B LLM that **trended #1 on Hugging Face**, a voice isolation model with the **best word error rate** on the Krisp benchmark and a paper on arXiv, and content at **600K monthly views**.

EXPERIENCE

Turnip · *Principal Software Engineer* Jan 2021 – Present

Four pivots in one startup, from game streaming to live audio to a creator social network to Zaps, an AI design suite.

- Built and ran the backend serving **25M users**, 20 microservices on Kubernetes exposed through Kong and talking internally over Kafka, with MongoDB as the primary store, Neo4j, Redis and Elasticsearch by use case, and real-time systems over MQTT. Sole backend engineer for the first two years.
- Wrote the live video streaming stack, **50K peak concurrent streams** and about **100TB** of video a day on RTMP and FFmpeg at **sub-2s latency**, holding sticky sessions over TCP through HAProxy ingress and packing streams onto shared Kubernetes pods for efficiency and cost.
- Implemented live audio rooms on Agora with MQTT holding presence and state for **2M monthly users** over **70M call minutes**.
- Designed the social network's feed and graph, weighting every interaction by a user and their friends into a per-user ranking cached and served at **sub-100ms**, with the graph on **Neo4j** powering second-degree friend discovery and suggestions.
- Shipped the AI editing suite of face swap, image enhancement and styles on **ComfyUI** with open-weights models, served on in-house GPU inference and then moved to a distributed GPU network, cutting GPU cost **80%** at about **2 seconds** an image and around **100K HD generations a day**.
- Replaced template tag search with semantic retrieval over **40K** embedded templates and carousels, served by vector search at **p95 under 50ms**.
- Set up an in-house AI agent harness for internal tooling where each project carries its own memory, TDD requirements, deployment skills and guardrails, a black box to the product team. **Ten non-engineers shipped 17 internal tools** with zero engineer commits.
- Migrated core services from Node to Go behind a strangler fig, with byte-parity responses under an enforced **100% test-coverage** gate.

MouseBelt · *Senior Backend Engineer (Contract)* Apr 2020 – Jan 2021

- Built a centralized cryptocurrency exchange end to end, accounts, KYC, markets and order flow on a **matching engine written in C** with gRPC and WebSocket layers.
- Ran custody across **six chains** (Bitcoin, Ethereum, EOS, Cosmos, ICON, BNB Chain) and built a multi-chain **DEX aggregator** with price scrapers and an on-chain indexer.

Paytm Insider · *Lead Backend Engineer* Jun 2017 – Apr 2020

India's biggest event ticketing platform.

- Led **seven engineers** across backend and mobile on the movie ticketing platform holding **500+ requests a second** on sale nights, where integrating Vista, the slow and error-prone legacy theatre software, meant redesigning the whole checkout from synchronous to asynchronous.

- Designed the logistics architecture, from internal tooling for the operations team to user-facing order tracking and delivery communication, integrated with logistics partners.
- Built the commerce flow end to end across cart, payments, inventory and taxation, **30 microservices** on EKS with SQS for async communication, Lambda and Kinesis.
- Moved on-demand workloads to serverless on Lambda and Kinesis, cutting cost by over **70%**.

SELECTED PROJECTS

All my projects, with live demos: arbazsiddiqui.me/projects

- **Ozan-v1-12B**: Open-weights 12B LLM for long-form fiction, fine-tuned with QLoRA and DPO on one A100 for about $40. Lowest slop of any 12B on EQ-Bench, ahead of Gemma 3 27B. **#1 trending on Hugging Face**, 3K+ monthly downloads. Hugging Face · Featherless
- **VoxLock**: Open-weights audio model for voice agents that isolates the caller's speech before speech-to-text. **Lowest word error rate on the Krisp benchmark** of every model tested, paid or open. 27M parameters, real-time on CPU with ONNX, Pipecat integration. arXiv paper · Demo · Hugging Face
- **sqlite-sparse**: Semantic search for RAG inside one SQLite file, a C extension with learned sparse vectors and no vector database or model at query time. **3.1 ms per query over 1M documents** against 582 ms for BM25 full-text search, also runs in the browser on WebAssembly. Demo · GitHub
- **The Footnote**: Autonomous AI video pipeline on open-weights LLMs, TTS and image generation, publishing at **600K monthly views** with no human in the loop. @thefootnotemedia
- **Iris**: A privacy-first cycle tracker, **4.9★** on the App Store in **twelve languages**, with Apple Health sync and partner sharing. Organic installs come from autonomous women's-health content pipelines on Instagram and Pinterest. App Store · Pinterest
- **lru-cache-node**: An O(1) LRU cache with expiry for Node.js, **119★**, featured in Node Weekly. GitHub
- **anabranch**: An HTTP load balancer and reverse proxy written in Go. GitHub

SKILLS

Languages	Go, TypeScript, JavaScript, Python
AI	RAG, vector search, learned sparse retrieval, in-browser inference (WebGPU, WebAssembly), speech and audio models (ASR, TTS, speech enhancement), agent orchestration (LangGraph, MCP, Claude Code), fine-tuning (QLoRA, DPO), model serving, GPU inference, evals, PyTorch, ONNX, Triton, MLflow, generative image pipelines (ComfyUI), diffusion models (Qwen, MiniMax, Flux), workflow automation (n8n), llama.cpp
Backend	Node.js, Kafka, REST, gRPC, WebSockets, microservices, event-driven architecture, RabbitMQ, SQS, Kinesis, MQTT, FastAPI, RTMP, FFmpeg
Data	MongoDB, PostgreSQL, Redis, Elasticsearch, Neo4j, Cassandra, Firebase RTDB
Infrastructure	Kubernetes (GKE, EKS), Docker, AWS, GCP, Terraform, Helm, Kong, Lambda, CI/CD
Observability	Prometheus, Grafana, ELK, OpenTelemetry, New Relic, Sentry
Crypto	Order matching engines, Multi-chain custody, DEX aggregation, NFT systems